

Simulation Object Boundary Module (SOBM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

Parent Standard: Simulation Object Standard (SOS)

Operational Layer: Simulation Object Governance Layer

Category: Governance & Enforcement

Subcategory: Simulation Governance

Type: Parent Standard Module

Governed Space: Simulation Object Boundary

Version: 1.0

Status: Canonical · Open Module

Effective Date: 15 August 2026

Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS® · INTEGROS® · ADIT® · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Governed Space

Simulation Object Boundary

Canonical Definition

Simulation Object Boundary Module (SOBM) defines the governance conditions under which the limits of an identified Simulation Object are formally established, distinguished from surrounding systems, environments, dependencies, actors, processes, interfaces, and external conditions, and maintained as a traceable Simulation Object Boundary throughout downstream simulation governance.

SOBM ensures that a simulation cannot silently expand, contract, or redefine what belongs to the Simulation Object after its identity has been established.

Operational Role

SOBM serves as the second internal governance space of the Simulation Object Standard (SOS).

SOIM establishes:

What exactly is the Simulation Object?

SOBM asks:

Where does that Simulation Object begin and end, and what lies outside it?

The governed progression is:

Identified Simulation Object → Boundary Context → Included Elements → Excluded Elements → Interfaces & Dependencies → Boundary Conditions → Boundary Status → Governed Simulation Object Boundary

SOBM establishes the structural perimeter within which subsequent object version, configuration, representation, assumptions, modeling, scenario, execution, and output governance can operate.

Module Operational Space

SOBM governs: Simulation Object boundary establishment, object inclusion, object exclusion, internal elements, external elements, system boundaries, subsystem boundaries, component boundaries, process boundaries, behavioral boundaries, organizational boundaries, spatial boundaries, temporal boundaries, functional boundaries, operational boundaries, environmental interfaces, external actors, external systems, dependencies, boundary-crossing interactions, boundary conditions, boundary assumptions, boundary ambiguity, boundary conflicts, boundary change, boundary expansion, boundary contraction, boundary inheritance, boundary traceability, and Simulation Object Boundary status.

Module Function

SOBM exists to prevent an identified Simulation Object from becoming structurally ambiguous once simulation design begins.

Its function is to establish which elements are:

Inside the Simulation Object

Outside the Simulation Object

and:

Connected Across the Simulation Object Boundary

This distinction matters because simulation conclusions may change materially depending upon what the simulation treats as part of the object itself versus part of its external environment.

SOBM therefore prevents: hidden boundary expansion, hidden boundary contraction, omitted material components, uncontrolled inclusion of external systems, environmental conditions being mistaken for object properties, dependencies being absorbed into the object without disclosure, object behavior being confused with external interaction, and simulation findings being generalized beyond the established object boundary.

Simulation Object Boundary

A Simulation Object Boundary is the governed perimeter separating the Simulation Object from the external conditions, entities, systems, processes, actors, and environments with which it may interact.

The boundary may be: physical, digital, logical, functional, organizational, behavioral, spatial, temporal, operational, or composite.

A single Simulation Object may therefore require several mutually consistent boundary dimensions.

Boundary Establishment

Once the Simulation Object has been identified through SOIM, its relevant boundaries must be established before downstream simulation structures depend upon them.

Boundary establishment should answer:

What belongs to the object?

What does not belong to the object?

Which external elements materially interact with it?

Through which interfaces do those interactions occur?

Which boundary conditions materially affect simulation behavior?

The required precision depends upon the intended simulation.

Included Elements

An Included Element is an element treated as part of the Simulation Object for the governed simulation.

Included elements may consist of: components, subsystems, software, hardware, internal processes, internal actors, internal control mechanisms, internal data structures, internal resources, or other constituent elements.

Inclusion must be sufficiently explicit where the element could materially affect simulation interpretation.

Excluded Elements

An Excluded Element is an element intentionally outside the Simulation Object Boundary.

Exclusion does not mean irrelevance.

An excluded element may still influence the object through an external interaction or dependency.

SOBM therefore distinguishes:

Outside the Object

from:

Irrelevant to the Simulation

These are not equivalent.

Boundary Interfaces

A Boundary Interface is a governed point through which the Simulation Object interacts with something outside its boundary.

Interfaces may include: data interfaces, physical connections, control interfaces, communication channels, human interactions, resource inputs, energy flows, sensor inputs, actuator outputs, organizational interactions, or environmental interactions.

Boundary interfaces should remain identifiable where their behavior materially affects simulation.

External Dependencies

A Simulation Object may depend upon entities that remain outside its boundary.

Examples may include: cloud infrastructure, navigation systems, external databases, communication networks, human operators, power systems, supply chains, regulatory inputs, external AI services, or supporting infrastructure.

SOBM requires these dependencies to remain distinguishable from the Simulation Object itself.

The existence of dependency does not automatically justify boundary inclusion.

Object Boundary vs Simulation Environment

SIMULOS® establishes a critical distinction:

Simulation Object Boundary ≠ Simulation Environment Boundary

The Simulation Object defines what is being simulated as the target.

The Simulation Environment defines the conditions within which simulation activity occurs.

For example:

Autonomous Drone

may be the Simulation Object.

Urban Airspace + Weather + Navigation Infrastructure + Other Aircraft

may form part of the simulated environment.

Those concepts must not be collapsed merely because they interact during simulation.

Object Boundary vs Reality Representation

SOBM establishes the boundary of the target object.

It does not determine whether that boundary has been represented accurately inside the simulation.

That later question belongs to the appropriate SIMULOS® Reality Representation governance space.

Therefore:

Object Boundary = What belongs to the target

while:

Representation Boundary = What the simulation representation actually models

A difference between the two may become a material simulation-governance issue.

Functional Boundary

A Simulation Object may require a functional boundary specifying which functions belong to the object being simulated.

For example, an autonomous vehicle simulation may include:

Perception

Planning

Decision

Vehicle Control

while an external navigation service remains outside the object.

The boundary should preserve this distinction even where the systems operate as one practical service.

Spatial Boundary

Where location or physical extent matters, the object may require a spatial boundary.

This may define: physical dimensions, geographic area, infrastructure extent, operational zone, or another spatial perimeter.

Spatial boundaries should not be confused with the wider simulation environment.

Temporal Boundary

A Simulation Object may also require a temporal boundary.

The object represented at:

Time T1

may not be materially equivalent to the same nominal object at:

Time T2

where its relevant characteristics have changed.

Detailed version governance belongs to the appropriate SOS module, but SOBM preserves the temporal perimeter relevant to the object definition.

Operational Boundary

An operational boundary identifies which operational functions, modes, states, or interactions belong to the Simulation Object for the particular simulation.

An object may have broader real-world capability than the operational portion included within a simulation.

SOBM requires this difference to remain explicit.

Composite Object Boundary

Composite Simulation Objects require governance over the relationship between:

Composite Boundary

and:

Constituent Boundaries

For example:

Autonomous Warehouse System

may contain:

Robots + Control System + Internal Network + Charging Infrastructure

while:

External Electrical Grid

remains outside the object boundary.

SOBM preserves both the composite perimeter and relevant internal constituent boundaries.

Nested Boundaries

Simulation Objects may contain nested boundaries.

Conceptually:

System Boundary

↓

Subsystem Boundary

↓

Component Boundary

↓

Functional Boundary

SOBM allows these boundaries to coexist where needed.

A lower-level simulation claim must not automatically be represented as applying to every higher-level boundary.

Boundary-Crossing Interaction

A Boundary-Crossing Interaction occurs whenever information, resources, commands, forces, signals, behaviors, or effects pass between the Simulation Object and an external element.

Such interactions may materially influence simulation behavior even though the external element remains outside the Simulation Object.

SOBM therefore requires material boundary-crossing interactions to remain visible.

Boundary Conditions

A Simulation Object Boundary Condition defines a governed condition applying at the interface between the object and its surroundings.

Examples may include: incoming data characteristics, resource availability, communication latency, physical forces, environmental exposure, human commands, external control signals, or other interface conditions.

Boundary conditions must not silently become assumed constants where variability is material.

Boundary Assumptions

Where a boundary cannot be fully established from known information, a boundary assumption may be necessary.

SOBM requires such assumptions to remain explicit.

Detailed governance of the assumption itself belongs to the SIMULOS® assumption-governance space.

SOBM preserves the fact that the object boundary depends upon that assumption.

Boundary Ambiguity

Simulation Object Boundary Ambiguity exists where it is unclear whether a materially relevant element belongs inside or outside the Simulation Object.

Ambiguity may arise from: incomplete architecture records, unclear ownership, dynamic system composition, external services, shared infrastructure, distributed control, unclear subsystem relationships, or evolving operational configurations.

Material ambiguity should not be silently resolved.

Boundary Conflict

A Boundary Conflict exists where credible sources define materially different boundaries for the same Simulation Object.

For example:

one system architecture may include an external decision service within the system boundary,

while another treats it as an external dependency.

Where the difference can affect simulation design or interpretation, the conflict must remain visible until governed.

No Silent Boundary Expansion Rule

SIMULOS® establishes:

A Simulation Object Boundary must not be expanded after simulation activity begins without explicit governance of the change.

Adding new components, functions, populations, dependencies, or operational

modes may materially change the Simulation Object.

No Silent Boundary Contraction Rule

Likewise:

Material elements may not be removed from the Simulation Object Boundary without explicit governance of the exclusion.

A simulation must not obtain more favorable results by quietly removing difficult or high-risk portions of the object.

No Environment Absorption Rule

An environmental element does not become part of the Simulation Object merely because it materially affects object behavior.

This rule protects the distinction between:

Object

and:

Environment

throughout the simulation lifecycle.

No Dependency Absorption Rule

A dependency does not automatically become part of the Simulation Object merely because the object cannot operate without it.

Dependency relationships must be visible.

Boundary inclusion requires an explicit basis.

No Boundary Inflation Rule

A simulation result established within one object boundary must not automatically be represented as applying to a broader boundary.

Therefore:

Subsystem Simulation ≠ Whole-System Simulation

Component Simulation ≠ Product Simulation

Single Operational Mode ≠ Complete Operational Capability

unless a governed basis supports the broader claim.

Boundary Change

A Simulation Object Boundary Change occurs when the governed perimeter of the object changes materially.

Possible changes include:

Expansion

Contraction

Recomposition

Interface Change

Dependency Change

or:

Boundary Reclassification

A boundary change may require downstream simulation governance to be reassessed.

Boundary Change Propagation

A material boundary change may affect: object version, configuration, reality representation, assumptions, model structure, input requirements, scenarios, execution, outputs, reality correlation, or simulation reliance.

SOBM therefore preserves boundary-change information for downstream governance rather than determining all downstream consequences itself.

Boundary Continuity

Where the Simulation Object remains in use across multiple simulations, its boundary should remain traceable between simulation cycles.

This enables comparison between:

Previous Simulation Object Boundary

and:

Current Simulation Object Boundary

and prevents materially different simulation scopes from being treated as equivalent.

Boundary Equivalence

Two simulations may use different technical representations while still concern the same Simulation Object Boundary.

Conversely, two simulations using the same model may concern materially different object boundaries.

SOBM therefore does not infer boundary equivalence from model similarity alone.

Simulation Object Boundary Status

Possible boundary statuses may include:

Established

Conditionally Established

Partially Established

Boundary Conflicted

Boundary Ambiguous

or:

Boundary Undetermined

Boundary status describes the strength of the object-boundary basis.

It does not establish simulation validity or operational reliance.

Simulation Object Boundary Record

SOBM should establish a persistent Simulation Object Boundary Record sufficient to reconstruct what was considered inside, outside, and connected to the Simulation Object when downstream simulation governance occurred.

The record may preserve: Simulation Object Identifier, boundary definition, boundary dimensions, included elements, excluded elements, constituent boundaries, external dependencies, boundary interfaces, boundary-crossing interactions, boundary conditions, boundary assumptions, ambiguity, conflicts, boundary status, boundary changes, and boundary traceability.

Failure Conditions

SOBM governance failure may exist where: the Simulation Object has no sufficiently defined boundary, material included elements are unidentified, material exclusions remain hidden, object and environment are improperly collapsed, dependencies are silently absorbed, material interfaces remain unidentified, boundary-crossing interactions are omitted, boundary assumptions remain hidden, conflicting boundaries are ignored, the object boundary is silently expanded or contracted, subsystem results are represented as whole-system results without sufficient basis, or downstream simulation activity cannot be traced to the boundary under which it was performed.

Minimum Implementation Framework

1. Establish the Candidate Object Boundary

Determine the physical, digital, logical, functional, spatial, temporal, operational, organizational, or composite perimeter required for the identified Simulation Object. 2. Classify Included, Excluded, and External Elements

Establish which material elements belong inside the object, which are explicitly excluded, and which remain external but relevant. 3. Identify Interfaces, Dependencies, and Boundary Conditions

Map material relationships crossing the Simulation Object Boundary and preserve conditions capable of affecting simulation behavior. 4. Resolve or Record Boundary Ambiguity

Identify conflicts, assumptions, uncertain inclusions, exclusions, and boundary limitations and assign the appropriate Simulation Object Boundary Status. 5. Establish the Simulation Object Boundary Record

Preserve the governed boundary, material elements, interfaces, dependencies, conditions, limitations, status, changes, and traceability required for downstream simulation governance.

Governance Outputs

SOBM may produce: Simulation Object Boundary Record, Object Boundary Definition, Boundary Dimension Record, Included Element Register, Excluded Element Register, External Element Register, Composite Object Boundary Record, Constituent Boundary Map, Boundary Interface Register, External Dependency Register, Boundary-Crossing Interaction Map, Boundary Condition Record, Boundary Assumption Reference, Boundary Ambiguity Record, Boundary Conflict Record, Simulation Object Boundary Status, Boundary Change Record, Boundary Expansion Record, Boundary Contraction Record, Boundary Continuity Record, Boundary Equivalence Assessment, and Simulation Object Boundary Traceability Map.

Use Case 1 — Spacecraft Mission Simulation

Scenario

A space institution simulates a spacecraft operating during a lunar mission.

The spacecraft depends upon: ground-control infrastructure, navigation data, communication networks, and mission-support systems.

Application

SOBM establishes the spacecraft and its defined onboard systems as the Simulation Object.

Ground control, external communication infrastructure, lunar environment, and navigation services remain outside the Simulation Object Boundary while their material interfaces and dependencies remain explicitly represented. Result

Simulation findings cannot silently treat the entire mission ecosystem as if it were part of the spacecraft itself.

The distinction between:

Spacecraft

External Infrastructure

and:

Simulation Environment

remains traceable throughout downstream SIMULOS® governance. Use Case 2 — Autonomous Industrial Facility

Scenario

An organization simulates an autonomous warehouse containing robots, control software, internal communications, charging systems, human workers, and external cloud services. Application

SOBM establishes which elements form the governed Simulation Object.

The autonomous warehouse control system and designated robotic fleet may fall inside the object boundary.

Human workers and external cloud services may remain outside the boundary while their interactions with the system remain material simulation inputs and interfaces. Result

The simulation cannot later be represented as covering the complete warehouse ecosystem if only the autonomous control system and robotic fleet were actually within the governed Simulation Object Boundary.

Architectural Position

Simulation Object Boundary Module (SOBM) is the second internal governance space of the Simulation Object Standard (SOS).

The internal progression begins:

Simulation Object Identification → Simulation Object Boundary

SOIM establishes:

Which object is being simulated?

SOBM establishes:

What belongs to that object, what remains outside it, and where does the object end?

Subsequent SOS modules may then govern the object's version, configuration, and documentation against a sufficiently identified and bounded Simulation Object.

SOBM does not replace governance of general governance boundaries under GOA™, operational boundaries under ORA™, validation scope under VALIDOS®, integrity under INTEGROS®, or autonomous-system constraints under ASGA™.

It governs specifically:

the boundary of the object that a simulation claims to represent.

Foundational Principle

A simulation claim cannot legitimately extend beyond the Simulation Object Boundary under which the simulation was governed and executed.

Canonical Closing Statement

Simulation Object Boundary Module (SOBM) establishes the structural perimeter of the Simulation Object by governing what belongs to the object, what remains outside it, and how material interfaces, dependencies, interactions, conditions, ambiguities, and boundary changes remain visible throughout the simulation lifecycle. Through explicit inclusion and exclusion, object-environment separation, interface and dependency mapping, anti-expansion and anti-contraction rules, boundary-status governance, change propagation, continuity, and traceability, SOBM prevents simulation results from acquiring broader object coverage than the boundary actually represented. By establishing a persistent Simulation Object Boundary before downstream version, configuration, representation, modeling, scenario, execution, and output governance proceeds, SOBM provides the second foundational object-governance mechanism of SIMULOS®.

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>